

Qualcomm AI Engine Direct (QAIRT) C++ API & Hexagon HTP v79 FastRPC Bindings

1. System Overview & Hardware Target

This specification details the native C/C++ interface to the Qualcomm AI Engine Direct (QAIRT / QNN SDK) targeting the Hexagon Tensor Processor (HTP v79 on Snapdragon 8 Elite SM8750).

2. Low-Level FastRPC Communication

The Hexagon NPU is interfaced via Qualcomm's Compute DSP Remote Procedure Call (`cdsprpc`) over Domain 3:

- Device Node: `/dev/fastrpc-cdsp`
- Userspace Driver: `libcdsprpc.so` / `libadsprpc.so`
- Execution Domain: User Process Domain (PD) on Domain 3 with attribute `0x8`
- Skel Library: `libQnnHtpV79Skel.so` loaded into DSP memory space
- Stub Library: `libQnnHtpV79Stub.so` linked into native Android C++ process

3. High-Speed Zero-Copy Shared Memory (ASharedMemory)

To bypass Android Binder 1MB transaction limits and eliminate CPU copying taxes:

- Allocate native anonymous shared memory: `ASharedMemory_create(name, size)`
- Map into local daemon address space: `mmap(NULL, size, PROT_READ | PROT_WRITE, MAP_SHARED, fd, 0)`
- Protect region across child processes: `ASharedMemory_setProt(fd, PROT_READ)`
- Transmit anonymous file descriptor over abstract UNIX domain socket (`sun_path[0] = '\0'`) using POSIX ancillary payloads (`SCM_RIGHTS`).

4. Multi-Device Virtualization & Process Domain Management

Qualcomm's mobile firmware enforces an allocation ceiling of ~3.5GB per single Hexagon Process Domain (PD). To execute models exceeding 3.5GB (such as Qwen 3.5 9B Q4_0 requiring ~5.5GB–6.0GB VRAM with KV cache):

- Allocate an array of QNN context handles: `Qnn_ContextHandle_t ctx[2]`
- Virtual Device 1 (`HTP0`): Maps Layers 0 to 24 (capped at ~2.8GB)
- Virtual Device 2 (`HTP1`): Maps Layers 25 to 48 (capped at ~2.7GB)

- Intermediate tensor transitions between Layer 24 and Layer 25 occur over on-chip FastRPC ioctl calls without host memory round-trips.